

Name: _____

Teachers Name: _____

SYDNEY TECHNICAL HIGH SCHOOL



Year 12

Mathematics Extension 1

Assessment Task 2

March 2014

Instructions

- Attempt all questions.
- Answers to be written on the paper provided.
- Start each question on a new page.
- All working must be shown. Full marks may not be awarded for poorly set out work.
- Indicated marks are a guide and may be changed slightly if necessary during the marking process.
- Approved calculators may be used.
- These questions must be handed in on top of your solutions.

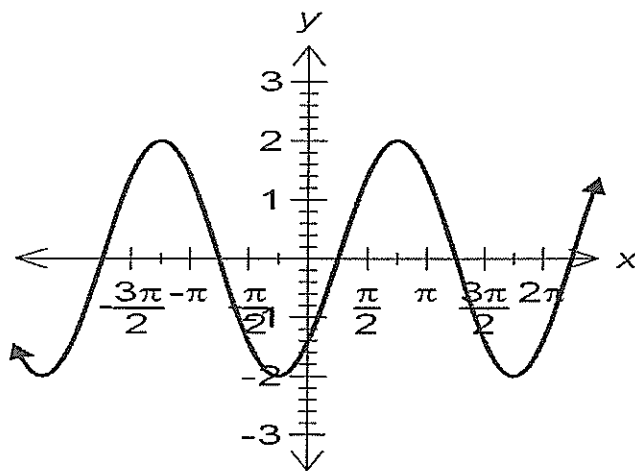
SECTION 1: MULTIPLE CHOICE:

Answer on the Multiple Choice Answer Sheet provided.

1. For what values of x is the curve $y = 4x^2 - x^3$ concave upwards?

A. $0 < x < \frac{8}{3}$ B. $x < \frac{4}{3}$
C. $x > \frac{4}{3}$ D. $x < 0, x > \frac{8}{3}$

2. Below is the graph of the function $y = f(x)$.



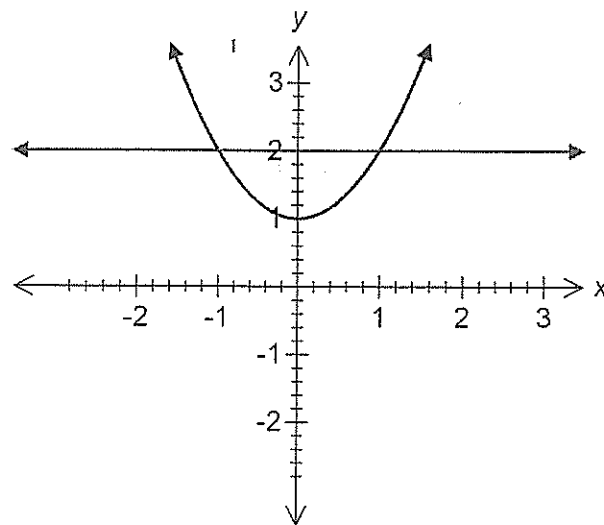
A possible equation for this function is:

A. $y = 2\sin(x + \frac{\pi}{4})$ B. $y = \sin 2(x - \frac{\pi}{4})$
C. $y = -2\cos(x + \frac{\pi}{4})$ D. $y = 2\sin(x - \frac{\pi}{4})$

3. $\lim_{x \rightarrow 0} \frac{\sin x}{7x} =$

A. 7 B. 0 C. $\frac{1}{7}$ D. ∞

4. The shaded area in square units is:



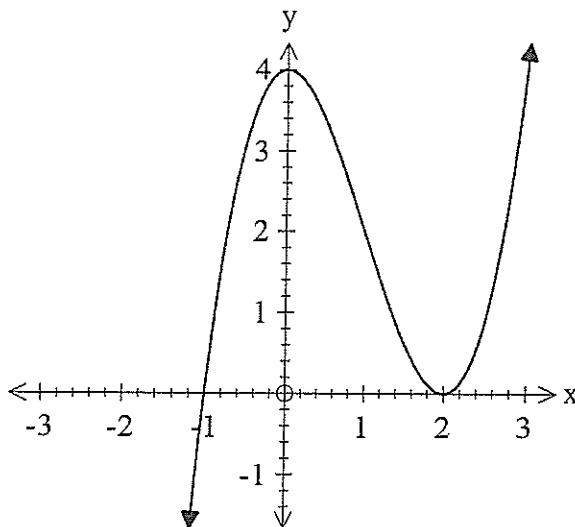
A. $\frac{4}{3}$

B. $\frac{10}{3}$

C. $\frac{2}{3}$

D. $\frac{20}{3}$

5. Below is a graph of $y = f(x)$.



Which of the following is true about $y = f'(x)$?

- a) There is a stationary point at $x = -1$.
- b) There is an inflexion point at $x = 1$.
- c) The x intercepts are -1 and 2.
- d) The y intercept is 0

SECTION 2: Write your solutions in the answer booklet provided.

QUESTION 6 (8 marks) **MARKS**

- a) Differentiate $\tan^2 x$ with respect to x . 1
- b) Find the exact value of the gradient of the tangent to the curve $y = x \sin x$ at the point where $x = \frac{\pi}{3}$. 2
- c) Find i) $\int \frac{x^2+3x+2}{x+1} dx$ 2
- ii) $\int_1^2 \frac{1}{2\sqrt{3x-2}} dx$ 3

QUESTION 7 **Start a new page** (8 marks)

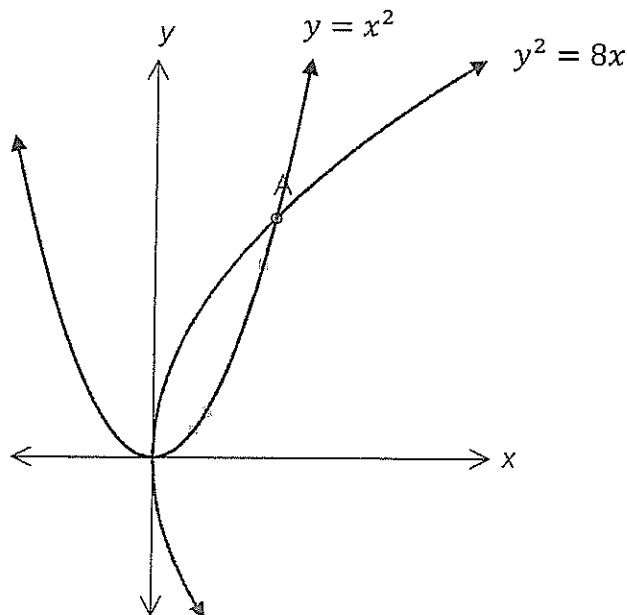
- a) The normal to the curve $y = 3 \tan 2x$ at the point P $(\frac{\pi}{8}, 3)$ cuts the y axis at Q. Find the coordinates of Q in exact form. 3
- b) A rectangle is cut from a circular disc of radius 3 cm.
- i) Show that the area of the rectangle is given by $A = l \sqrt{36 - l^2}$, where l is the length of the rectangle. 2
- ii) Find the area of the largest rectangle that can be produced. 3

QUESTION 8**Start a new page** (8 marks)**MARKS**

- a) Calculate the area bounded by the curve $y = x^2 - 4$ and the x axis, between $x = 1$ and $x = 3$. 3
- b) i) Sketch $y = 3\sin 2x$ for $0 \leq x \leq \pi$. 1
- ii) State the period and amplitude of $y = 3\sin 2x$. 2
- iii) On the same set of axes, sketch $y = \cos x$ for $0 \leq x \leq \pi$. 1
- iv) State how many solutions exist for $3\sin 2x - \cos x = 0$ for $0 \leq x \leq \pi$. 1

QUESTION 9**Start a new page** (8 marks)

- a) $\int 5 \sin(3x + 2) dx$ 1
- b) Using the substitution $u = 2 - x$, or otherwise, evaluate $\int_1^2 x\sqrt{2-x} dx$. 3
- c) The curves $y^2 = 8x$ and $y = x^2$ intersect at A.



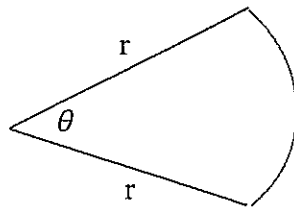
- i) Find the coordinates of A. 1
- ii) Calculate the volume of the solid generated when the shaded region is rotated about the y axis. 3

QUESTION 10**Start a new page** (8 marks)**MARKS**

- a) i) $\int_{-5}^5 \sqrt{25 - x^2} dx$ represents the area of a semi-circle. 1
Calculate the exact area of this semi-circle.
- ii) Use Simpson's Rule with 5 function values to find an approximation to $\int_{-5}^5 \sqrt{25 - x^2} dx$, correct to 3 decimal places. 2
- iii) By comparing your results from parts i) and ii), find the percentage error in the use of the Simpson's Rule for the approximation of the actual area. 2
- b) The area enclosed between the curve $y = 2\sin x$ and the x axis for $0 \leq x \leq \pi$ is rotated about the x axis. Find the volume of the solid formed. 3

QUESTION 11**Start a new page** (8 marks)

a)



The diagram shows the section of a circle of radius r metres and angle θ radians.

- If the area of the sector is $50m^2$, find an expression for the perimeter of the sector in terms of r only. 2
- b) For the curve $y = x^4 - 2x^3$,
- i) Find the stationary points and determine their nature. 3
- ii) Find any inflexion points. 2
- iii) Sketch its graph, showing all important features. 1

END OF TEST

YEAR 12 MATHEMATICS EXTENSION 2 TASK 2 2014

SECTION 1

1. D
2. D
3. C
4. A
5. D

SECTION 2

QUESTION 6 8 marks

a) $\frac{d}{dx} \tan^2 x = 2 \tan x \sec^2 x$ ①

b) $y = x \sin x$
 $\frac{dy}{dx} = \sin x + x \cos x$ ①
 at $x = \frac{\pi}{3}$, $\frac{dy}{dx} = \sin \frac{\pi}{3} + \frac{\pi}{3} \cos \frac{\pi}{3}$
 $= \frac{\sqrt{3}}{2} + \frac{\pi}{3} \times \frac{1}{2}$
 $= \frac{2\sqrt{3} + \pi}{6}$ ①

c) i) $\int \frac{x^2 + 3x + 2}{x+1} dx$
 $= \int \frac{(x+1)(x+2)}{x+1} dx$
 $= \int (x+2) dx$
 $= \frac{1}{2}x^2 + 2x + C$ ②
 (must have 'C')

ii) $\int_1^2 \frac{1}{2\sqrt{3x-2}} dx$
 $= \frac{1}{2} \int_1^2 (3x-2)^{-\frac{1}{2}} dx$ ①
 $= \frac{1}{2} \left[\frac{2}{3} (3x-2)^{\frac{1}{2}} \right]_1^2$ ①
 $= \frac{1}{3} (4^{\frac{1}{2}} - 1^{\frac{1}{2}})$
 $= \frac{1}{3}$ ①

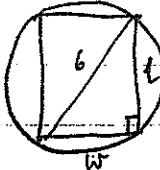
QUESTION 7 8 marks

a) $y = 3 \tan 2x$
 $\frac{dy}{dx} = 6 \sec^2 2x$
 at $x = \frac{\pi}{8}$, $\frac{dy}{dx} = 6 \sec^2 \frac{\pi}{4}$ ①
 $= 12$

$\therefore m$ of tangent = 12
 m of normal = $-\frac{1}{12}$ (for perpendicular lines $m_1 m_2 = -1$)

$y - 3 = -\frac{1}{12}(x - \frac{\pi}{8})$ ①
 $12y + x - 36 - \frac{\pi}{8} = 0$

at $x = 0$ $y = 3 + \frac{\pi}{96}$
 $\therefore$ co-ords of Q are $(0, 3 + \frac{\pi}{96})$ ①

b)  i) $6^2 = l^2 + w^2$ (Pythagoras Theorem)
 $w = \sqrt{36 - l^2}$ ①

$A = lw$
 $= l\sqrt{36 - l^2}$ ①

ii) $\frac{dA}{dl} = (36 - l^2)^{\frac{1}{2}} + l \times \frac{1}{2} (36 - l^2)^{-\frac{1}{2}} \times -2l$
 $= (36 - l^2)^{\frac{1}{2}} (36 - l^2) - l^2$
 $= \frac{36 - 2l^2}{\sqrt{36 - l^2}}$ ①

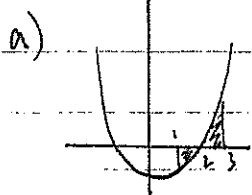
$\frac{d^2A}{dl^2} = \frac{-4l\sqrt{36 - l^2} - \frac{1}{2}(36 - l^2)^{-\frac{1}{2}} \times -2l(36 - 2l^2)}{36 - l^2}$
 $= \frac{l(36 - l^2)^{\frac{1}{2}} (-4(36 - l^2) + 36 - 2l^2)}{36 - l^2}$
 $= \frac{l(-108 + 2l^2)}{36 - l^2}$ ①

stat pts at $\frac{dA}{dl} = 0$
 $36 - 2l^2 = 0$
 $l = \pm 3\sqrt{2}$ but l is +ve
 $A = 3\sqrt{2} \sqrt{36 - 18}$
 $= 18$

at $l = 3\sqrt{2}$, $\frac{d^2A}{dl^2} < 0 \therefore$ Maximum ①

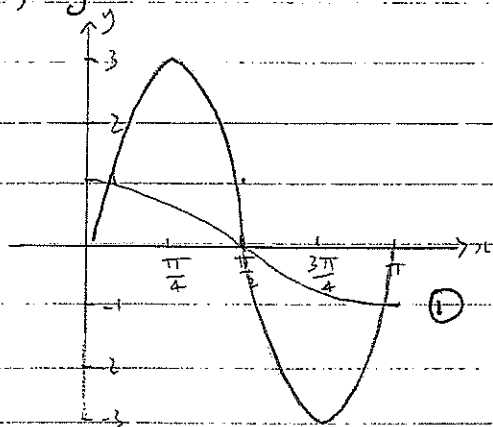
$\therefore$ Maximum area is 18 units²

QUESTION 8 8 marks



$$\begin{aligned}
 A &= \left| \int_{-2}^0 (x^2 - 4) dx \right| + \int_0^3 (x^2 - 4) dx \quad (1) \\
 &= \left| \left[\frac{1}{3}x^3 - 4x \right]_{-2}^0 \right| + \left[\frac{1}{3}x^3 - 4x \right]_0^3 \quad (1) \\
 &= \left| \left(\frac{8}{3} - 8 \right) - \left(\frac{1}{3} - 4 \right) \right| + (9 - 12) - \left(\frac{8}{3} - 8 \right) \\
 &= 4 \text{ units}^2 \quad (1)
 \end{aligned}$$

b) i) $y = 3 \sin 2x$



- ii) Period = π , amplitude = 3 (1)
 iii) on graph
 iv) $3 \sin 2x = \cos x$
 3 solutions (1)

QUESTION 9 8 marks

a) $-\frac{8}{3} \cos(3x+2) + c$ (1)

(Don't deduct for no 'c')

b) $u = 2 - x \rightarrow x = 2 - u$

$du = -dx$

at $x=1, u=1$
 $x=2, u=0$

$$\begin{aligned}
 \int_1^2 x \sqrt{2-x} dx &= \int_1^0 (2-u) \sqrt{u} (-du) \quad (1) \\
 &= \int_0^1 (2u^{\frac{1}{2}} - u^{\frac{3}{2}}) du \\
 &= \left[\frac{4}{3} u^{\frac{3}{2}} - \frac{2}{5} u^{\frac{5}{2}} \right]_0^1 \\
 &= \frac{4}{3} - \frac{2}{5} \\
 &= \frac{14}{15} \quad (1)
 \end{aligned}$$

c) i) $x^4 = 8x$

$x(x^3 - 8) = 0$

$x=0, x=2$

$y=0, y=4$

$\therefore$ Co-ords of A are (2, 4) (1)

ii) $V = \pi \int_0^4 \left(y - \frac{y^4}{64} \right) dy$ (1)

$= \pi \left[\frac{1}{2} y^2 - \frac{y^5}{320} \right]_0^4$ (1)

$= \pi \left(8 - \frac{16}{5} \right)$

$= \frac{24\pi}{5} \text{ units}^3$ (1)

QUESTION 10 8 marks

a) i) $A = \frac{1}{2} \times \pi \times 5^2$ (1)
 $= \frac{25\pi}{2} \text{ units}^2$

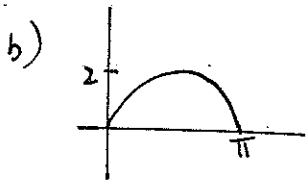
ii)

x	-5	$-2\frac{1}{2}$	0	$2\frac{1}{2}$	5
y	0	$\frac{5\sqrt{3}}{2}$	5	$\frac{5\sqrt{3}}{2}$	0

$$\int_{-5}^5 \sqrt{25 - x^2} dx$$

$$\doteq \frac{2\frac{1}{2}}{3} (0 + 0 + 2(5) + 4(\frac{5\sqrt{3}}{2} + \frac{5\sqrt{3}}{2}))$$
 (1)
$$\doteq 37.201 \text{ units}^2 \text{ (3 dp)}$$
 (1)

iii) $\frac{\text{ii) - i)}}{\text{i)}} \times 100 = 5.269\% \text{ (3 dp)}$ (2)



$$V = \pi \int_0^\pi 4 \sin^2 x dx$$
 (1)
$$= 4\pi \int_0^\pi \frac{1}{2} (1 - \cos 2x) dx$$

$$= 2\pi \int_0^\pi [x - \frac{1}{2} \sin 2x]_0^\pi$$
 (1)
$$= 2\pi (\pi - \frac{1}{2} \sin 2\pi - 0)$$

$$= 2\pi^2 \text{ units}^3$$
 (1)

QUESTION 11 8 marks

a) $A = \frac{1}{2} r^2 \theta = 50$
 $\theta = \frac{100}{r^2}$

$$l = r\theta$$

$$= r \times \frac{100}{r^2}$$
 (1)
$$= \frac{100}{r}$$

$$P = 2r + \frac{100}{r}$$
 (1)

b) i) $y = x^4 - 2x^3$
 $\frac{dy}{dx} = 4x^3 - 6x^2$
 $\frac{d^2y}{dx^2} = 12x^2 - 12x$

stat pts at $\frac{dy}{dx} = 0$

$$4x^3 - 6x^2 = 0$$

$$2x^2(2x - 3) = 0$$

$$x = 0 \quad x = \frac{3}{2}$$
 (1)

$$y = 0 \quad y = -\frac{27}{16}$$

at $x = 0$, $\frac{d^2y}{dx^2} = 0$

x	$-\frac{1}{2}$	0	$\frac{1}{2}$
$\frac{d^2y}{dx^2}$	> 0	0	< 0

$\therefore$ concavity changes

$\therefore$ Horizontal inflexion pt at (0,0)

at $x = \frac{3}{2}$, $\frac{d^2y}{dx^2} > 0$ (1)

$\therefore$ concave upwards

$\therefore$ Minimum turning point at $(\frac{3}{2}, -\frac{27}{16})$

iii) For inflexion pts $\frac{d^2y}{dx^2} = 0$

$$12x^2 - 12x = 0$$

$$12x(x - 1) = 0$$

$$x = 0, \quad x = 1$$

$$y = 0 \quad y = -1$$
 (1)

See ii)

at $x = 1$:

x	$\frac{1}{2}$	1	$\frac{3}{2}$
$\frac{d^2y}{dx^2}$	< 0	0	> 0

$\therefore$ concavity changes

$\therefore$ Inflexion point at (1, -1)

